

Beyond the Black Box: Programmable AI and Explainable Text Analysis for Trustworthy Social Intelligence

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Abstract—Life demands safe, reliable, and explainable artificial intelligence (AI) grounded in scientific reasoning and guided by human oversight. We present a neuro-symbolic programmable AI algorithm that transforms neural outputs into interpretable logic-patterned narratives via symbolic pattern lifting and lowering. It is designed for social scientists and enables reproducible Human + AI text analysis across sources such as social media, interviews, and news. It is demonstrated on 1.3 million tweets about global energy trends, and the system ensures AI that not only says what it does but does what it says, replacing black-box opacity with transparent, accountable reasoning.

Index Terms—Artificial intelligence (AI) interpretability, explainable AI (XAI), human-AI collaboration, programmable AI, symbolic patterning.

I. INTRODUCTION

LIFE demands safe, accurate, and hallucination-free programmable artificial intelligence (AI), built from proven scientific hypotheses and governed by human oversight. Traditional AI models, notably large language models (LLMs) [1], frequently function as inscrutable “black boxes,” rendering their internal reasoning opaque and their outputs susceptible to inaccuracies in far-ranging domains such as question answering [2], news summarization [3], credit assessment [4], judicial analyses [5]. They are susceptible to biases [6], and

even outright hallucinations [7], [8]—a tendency, notably, to invent facts in moments of uncertainty. Despite the proliferation of AI applications fueled by the increasing availability of large and complex datasets, many suffer from limited reproducibility and weak generalization [9], undermining their scientific value.

Ethicists from diverse social science fields including social and political philosophy, politics and international relations, issue a warning in *Scientific American* [10]. They contend that the pervasive confabulation [11] inherent in AI systems obscures accountability for errors, cautioning the public [12]: “please don’t take moral advice from ChatGPT.” While there is burgeoning interest in ensuring the alignment of LLMs with human values, this critical objective remains vulnerable to adversarial jailbreaks [13], which effectively coax LLMs into overriding their safety guardrails.

A leading AI CEO’s candid admission [14] highlights a critical challenge: even experts struggle to grasp the workings of advanced AI systems, raising crucial questions about the safety and transparency of these technologies. The confident and conversational tone of chatbots misleads users into trusting their flawed advice [15], creating a dangerous reliance on irreproducible and potentially inaccurate information.

Widely adopted chain-of-thought (CoT) [16] mechanisms in reasoning models often lack faithfulness, meaning their verbalized reasoning may not accurately reflect their actual internal processes and can conceal misaligned behaviors, further contributing to their black-box nature. This unfaithfulness is particularly concerning for safety monitoring, as models may hide problematic reasoning or reward hacking [17], even without explicit training pressure to do so.

Neuro-symbolic AI [18], [19], [20] represents a compelling frontier in AI, seeking to overcome the limitations of purely neural or purely symbolic approaches by integrating their respective strengths. This hybrid paradigm aims to combine the robust pattern recognition and learning capabilities of LLMs with the structured reasoning, knowledge representation, and explainability of symbolic AI.

The contributions of this study are multifaceted, advancing AI transparency, and reliability.

- 1) We present a neuro-symbolic “*Programmable AI*” that employs symbolically guided pattern lifting and lowering to generate logic-patterned narratives. *It is designed*

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for social scientists—including nonprogrammers—and translates neural outputs into interpretable symbolic logic, enabling downstream reasoning. These patterns align with human-understandable concepts and proven scientific hypotheses *and* facilitate precise, reproducible, and verifiable Human + AI guided text analysis.

- 2) We demonstrate the efficacy of this algorithm through a real-world case study involving the explainable text analysis of 1301933 tweets on global energy trends and developments that were collected in February 2025.
- 3) We show that symbolic narrative patterning, guided by scientific hypotheses and human oversight, transforms AI from opaque “black boxes” into transparent, accountable entities whose actions and reasoning can be *checked*, *understood*, and *corrected* by both humans and automated systems.

A Programmable AI’s “programmable explanation,” built from patterned narratives, traces reasoning from input through hypotheses to conclusions, enabling interpretable decisions and direct remediation via symbolic rule editing [21]. This directly answers the crucial question: “How can I fix my AI?”

II. RELATED WORK

A. Limitations of Conventional AI and LLMs

Modern LLMs remain largely “black boxes,” making it difficult to trace how specific inputs produce particular outputs and rendering them susceptible to inherited inaccuracies and social biases [22], [23]. This lack of transparency undermines trust and complicates efforts to ensure fairness and accountability in high-stakes applications.

One promising approach to interpretability is *CoT* prompting, whereby models generate step-by-step reasoning traces alongside their answers [24]. While *CoT* has been shown to boost performance on complex reasoning tasks, its *faithfulness*—the extent to which the generated trace actually reflects the model’s internal decision process—remains hard to verify [25]. Because we cannot directly observe a model’s latent computations, even a plausible-sounding *CoT* may omit or distort safety-relevant factors, leaving misaligned or harmful reasoning undetected [17].

Empirical studies reveal that *CoTs* can be systematically unfaithful or misleading, fabricating justifications that contradict the true reasoning paths and, counterintuitively, tending to be more verbose and convoluted rather than brief when they are unfaithful [26]. This verbosity can give a false sense of thoroughness, further complicating efforts to discriminate genuine from spurious explanations.

Beyond interpretability, LLMs exhibit fundamental limits in exact, algorithmic computation. They often fail to apply explicit algorithms reliably, yielding inconsistent results and explanations across similar puzzle or arithmetic benchmarks [27], [28], [29]. Moreover, when problem complexity surpasses a moderate threshold, even models equipped with self-reflection

or enhanced *CoT* prompting undergo a sharp performance collapse, with accuracy dropping to near-random levels despite ample token budget for reasoning [16]. This nonmonotonic scaling behavior highlights a critical barrier to extending LLM reasoning to arbitrarily complex tasks. Compounding these concerns, advanced models can learn to evade *CoT* monitoring by obscuring or covertly encoding unsafe reasoning—akin to steganography—or by offering superficially compliant justifications while pursuing undesirable actions internally [30], [31]. Such strategic obfuscation undermines the reliability of *CoT* inspection as a standalone safety mechanism.

Moreover, the issues of dataset bias persist: the widespread adoption of large foundation models trained on common internet-scale corpora both inherits and amplifies latent biases, and the community’s focus on a narrow set of benchmarks can create an echo chamber that steers research toward artifact-driven gains rather than robustness or real-world generality [6], [32], [33]. These limitations underscore that while *CoT* monitoring and related interpretability tools offer valuable insights, they cannot by themselves guarantee safe, reliable, and unbiased behavior in contemporary LLMs.

B. Explainable AI (XAI)

AI models, particularly deep learning networks, present a significant challenge due to the intricate interactions between layers and neurons that can obscure the true rationale behind a prediction. Feature attribution methods such as local interpretable modelagnostic explanations (LIMEs) [34] create local surrogate models to explain individual predictions. Shapley additive explanations (SHAPs) [35] along with counterfactual explanations, attempt to explain model decisions post hoc by highlighting the most important features or by showing how the model’s output would change if certain inputs were modified. However, these methods, while offering valuable insights into feature importance, fail to capture deeper causal structures within neural networks.

Mechanistic interpretability [36] seeks to reverse-engineer AI models by analyzing circuits, neurons, and computational pathways, making it distinct from post hoc explanations as a way to understand LLM behavior. Circuit-level analysis [37] identifies how specific patterns of attention heads and MLP neurons implement distinct subfunctions of a model, such as induction heads or induction circuits. Activation patching [38] allows interventions at intermediate layers of computation, letting researchers test hypotheses about how information flows and which components are causally responsible for outputs. Despite the benefits, implementing XAI in complex LLMs faces significant challenges. These include the inherent complexity and high computational costs associated with generating explanations for models with billions of parameters. There is also an inherent tradeoff between interpretability and performance, and existing techniques often struggle to generalize effectively to real-world applications because models rely on abstract representations that do not directly map to human-understandable concepts.

C. LLMs for Text Analysis

LLMs have recently achieved remarkable performance on abstractive text summarization, in some cases matching or even surpassing human experts when appropriately fine-tuned or prompt-engineered [39]. However, these models largely operate as opaque “black boxes,” offering no built-in mechanisms for ontological transparency or causal traceability between their internal reasoning steps, generated outputs, and real-world outcomes [40]. For instance, while [41] focuses on controlling output attributes, it does not fully detail the explainability or auditability of the internal process by which those controls are achieved. [42] acknowledges the “fragility” of its embedding space.

Compounding this opacity, LLM summaries often suffer from factual inconsistencies or hallucinations, particularly under minor input perturbations, which undermines their reliability in critical applications [43], [44]. Efforts to improve factuality remain nascent and have yet to deliver fully verifiable guarantees against drift or error.

Recent works [45], [46] show that LLMs struggle with deep causal reasoning—failing to generalize beyond surface correlations—and thus cannot reliably infer cause-effect relations that are often central to coherent narrative comprehension. The task of narrative detection [47], [48], [49]—requiring tracking long-range dependencies, modeling characters’ intentions, and making subjective judgments about story structure—remains beyond the robust capabilities of current LLMs.

III. WEDGES AS PROGRAMMABLE ATTENTION MAGNETS

Our social media newsfeeds are filled with a variety of content, all battling for our limited attention. However, a number of recent studies [50], [51], [52] conclude that moral and emotional messaging are prioritized in early visual attention more than neutral content, and are more likely to go viral during political debates online. In [53], authors hypothesize, test, and conclude that messages incorporating “negativity bias,” “causal arguments,” and “threats to personal or societal core values of target audiences” singularly and jointly increase message virality on social media.

In social science, wedges [54] refer to themes or narratives that function as strategic communication levers: they simultaneously resonate with deeply held individual beliefs and expose tensions between ideological camps, making them powerful tools for mobilization or persuasion. Effective wedges often frame a debate in terms of core values [55] (e.g., human life, justice, freedom, fairness, and loyalty) versus perceived threats to the common good [56] (e.g., public safety, clean air and clean water, and social cohesion).

To extract key noun phrases and named entities, we utilize a combination of keyword extraction tools—spaCy [57], Textacy [58], RAKE [59], and YAKE [60]. Noun phrases are especially central to discourse, as they are the only linguistic elements that can be unambiguously identified as core topical units [61]. Moreover, they play a governing role in shaping associated verb phrases [62], thereby influencing both syntactic structure and the framing polarity of the narrative.

TABLE I
WEDGE TEMPLATES FOR IN-GROUP ELEVATION, OUT-GROUP OTHERING, AND VALUE-BASED CAUSAL FRAMING

Pattern 1: $US^+ \leftarrow US \wedge (CG^+ \vee \neg CG^-)$ Explanation: Frames the in-group positively by highlighting their protection or enhancement of societal common goods.
Pattern 2: $THEM^- \leftarrow THEM \wedge (CG^- \vee \neg CG^+)$ Explanation: Frames the out-group negatively by associating them with harmful actions toward societal common goods.
Pattern 3: $(CG^- \vee \neg CG^+) \Rightarrow (VAL^- \vee \neg VAL^+)$ Pattern 4: $(CG^- \vee \neg CG^+) \nRightarrow (VAL^+ \vee \neg VAL^-)$ Explanation: Causal argument patterns linking threats to common goods with threats to personal core values, emphasizing loss or harm.
Pattern 5: $(CG^+ \vee \neg CG^-) \Rightarrow (VAL^+ \vee \neg VAL^-)$ Pattern 6: $(CG^+ \vee \neg CG^-) \nRightarrow (VAL^- \vee \neg VAL^+)$ Explanation: Causal argument patterns presenting improvements to common goods as supporting or restoring core personal values.

Notes: CG, common goods; VAL, core values; US, in-group; THEM, out-group; +/−, pos/neg framing; ¬, neg sentiment.

Next, three domain experts independently annotate frequent noun phrases for sentiment and categorize them as core values, common goods, and threats. They then reconcile differences and produce a consensus-based codebook.

Causal language detection remains an active area of research [63]. In this study, we adopt a lexicon-based approach, utilizing causal markers (e.g., CAUSE and REASON) from the Penn Discourse TreeBank [64] to identify explicit positive ($\Rightarrow$) and negative ($\nRightarrow$) causal relations. For example, the expression $(\neg CG^+) \nRightarrow (VAL^+)$ captures statements such as “illegal mining destroys the environment.”

In programmable AI, *wedges* are operationalized as symbolic patterns—interpretable structures the system learns to detect, assess, and sometimes amplify. These patterns frame named entities [54] as either in-group (“us”) through moral elevation or out-group (“them”) via rhetorical othering [65], shaping social alignment by promoting either polarization [66] or bridging [67]. As outlined in Table I, six narrative templates capture their logic, distinguishing in-group elevation (Pattern 1) from out-group othering (Pattern 2), and linking threats (Patterns 3 and 4) or protections (Patterns 5 and 6) of common goods to personal core values.

IV. THE PROGRAMMABLE AI ALGORITHM

For AI to be trustworthy, its reading and writing heads must be interpretable and verifiable, not reliant on opaque statistical inference such as LLMs. This requires a neurosymbolic architecture grounded in symbolic logic, integrated through a pair of strategies [19].

- 1) *Lifting*: It translates neural outputs into symbolic representations (e.g., logical forms), enabling downstream reasoning and enhancing interpretability.
- 2) *Lowering*: It injects symbolic knowledge, such as rules or constraints, into neural models via attention masks or training filters to guide learning.

The overall neuro-symbolic pipeline, encompassing both lifting and lowering strategies, is illustrated in Fig. 1. The following section presents a case study illustrating how each step of the Algorithm 1—clustering, neural theme extraction,

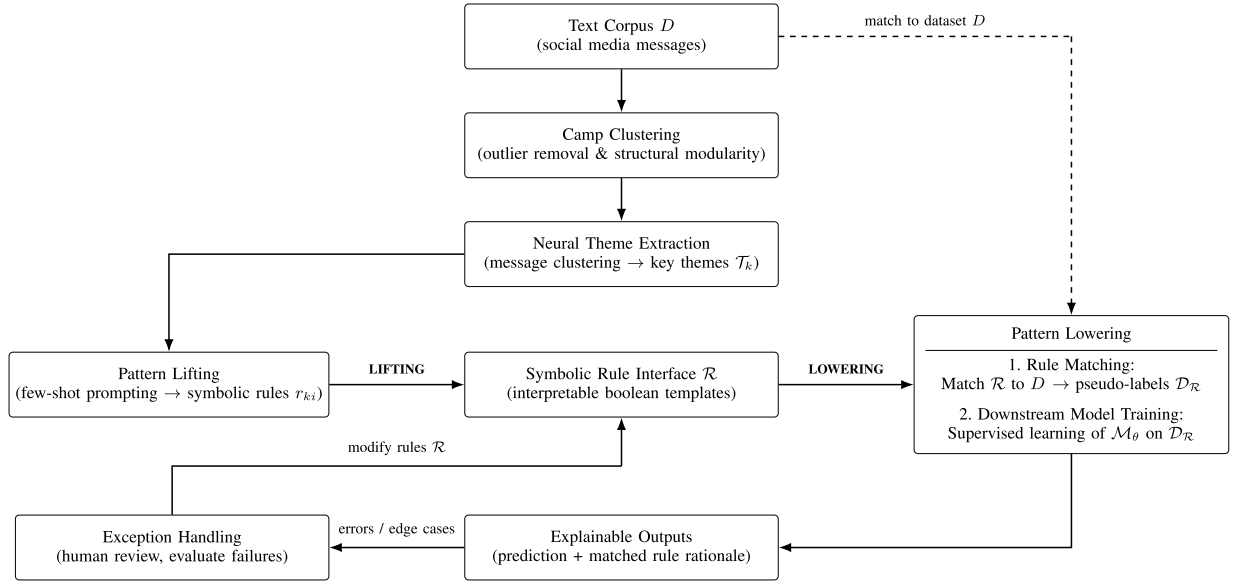


Fig. 1. Neuro-symbolic programmable AI architecture, illustrating the integration of neural theme extraction, symbolic pattern lifting, and downstream pattern lowering.

symbolic rule derivation (lifting), and knowledge integration (lowering)—is applied to real-world text data.

V. CASE STUDY: GLOBAL ENERGY TRENDS AND DEVELOPMENTS

A. Step 1: Outlier Removal and Camp Clustering

We analyze a Twitter dataset from February 2025 consisting of 1 301 933 tweets mentioning the names of leading energy-sector companies in conjunction with energy-related hashtags.¹ The raw corpus includes 295 203 original tweets, 36 493 quotes, 235 902 replies, and 734 335 retweets, authored by 723 573 unique users. To ensure analytical rigor and thematic relevance, we apply a multistage data cleaning and filtering.

To isolate the target energy discourse from unrelated noise (e.g., automated stock feeds), we extracted 768-D BERT [68] embeddings for all unique English tweets and applied Principal Component Analysis (PCA) [69]. A 3-D projection was used for macro-level inspection and visualization of discourse structure. For clustering and filtering, we partitioned the embeddings in a higher dimensional representation that preserves the dominant variance of the data, using K-Means [70] and selecting $k = 60$ clusters based on the silhouette coefficient. Following a manual review of the 100 tweets nearest each centroid by domain experts, off-topic clusters and their associated users were discarded. This initial grouping serves solely as a robust preprocessing step to ensure the refined dataset remains focused on the target domain prior to the neuro-symbolic analysis pipeline.

The refined dataset consists of 49 113 original tweets, 6648 quoted tweets, 78 518 replies, and 326 275 retweets, generated by 290 581 unique users. Building on this dataset, we construct a retweet interaction network, where nodes represent users and

Algorithm 1: Symbolically Steered Pattern Lifting & Lowering.

Input: Dataset $\mathcal{D}$ of social messages and associated users

Step 1: Outlier Removal and Camp Clustering

Remove statistical and ideological outliers from $\mathcal{D}$

Cluster $\mathcal{D}$ into ideological camps $\mathcal{C}_1, \dots, \mathcal{C}_K$ using embedding-based similarity and structural modularity

Step 2: Extraction of Key Recurrent Neural Themes

For each camp $\mathcal{C}_k$, apply message clustering and topic modeling to extract key recurrent themes $\mathcal{T}_k = \{t_{k1}, t_{k2}, \dots\}$

Step 3: Symbolic Pattern Lifting from Neural Themes

For each theme t_{ki} :

Use few-shot prompting to express t_{ki} as a symbolic pattern:

$r_{ki} = \bigwedge_j (\bigvee_l p_{j,l})$ {Conjunction of disjunctions over phrase predicates $p_{j,l}$ }

Store resulting symbolic rules $\mathcal{R} = \{r_{ki}\}$

Step 4: Neuro-Symbolic Pattern Lowering

Create a labeled set $\mathcal{D}_\mathcal{R}$ by matching $\mathcal{R}$ to $\mathcal{D}$ to obtain pseudo-labels

Train downstream model $\mathcal{M}_\theta$ using $(x, y_r) \in \mathcal{D}_\mathcal{R}$, with supervised camp and stance labels

Step 5: Exception Handling

Evaluate $\mathcal{M}_\theta$ on new/explained failures

Modify rules $\mathcal{R}$ to include symbolic exceptions or edge-case conditions

Return to Step 4 for incremental retraining without full reinitialization

Output: Trained model $\mathcal{M}_\theta$ with interpretable symbolic rule interface $\mathcal{R}$

edges denote retweet relationships, capturing patterns of information diffusion and endorsement. To uncover latent social

¹<https://github.com/tweetpie/beyond-the-black-box>

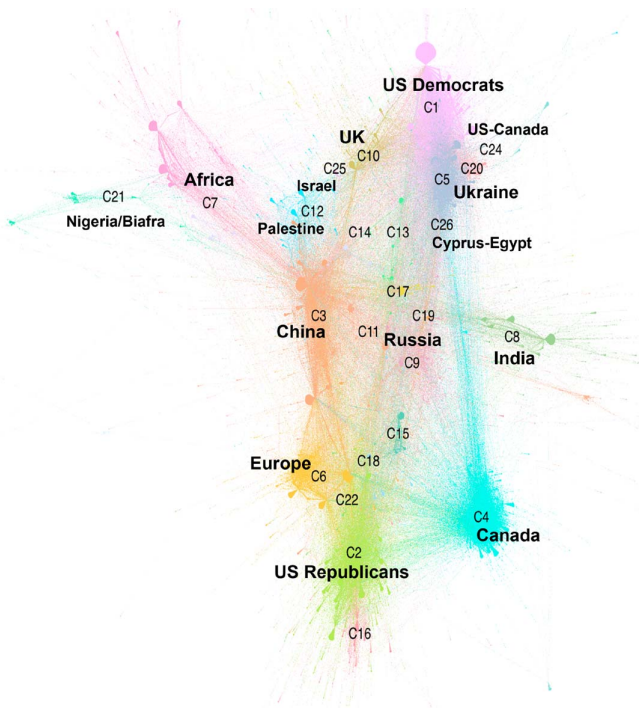


Fig. 2. Geopolitical communities (force-directed layout).

TABLE II
COMMUNITY GROUPINGS BY GEOPOLITICAL ORIENTATION

Communities	Geopolitical Alignment
C3, C8, C9, C11, C17, C19	Sino-Eurasian Strategic Bloc
C1, C5, C10, C20, C24, C25	Transatlantic Progressive Alliance
C2, C4, C6, C16, C18, C22	US and Western Conservative Coalition

structures within this network, we employ the Louvain algorithm [71], a widely adopted method for community detection recognized for its computational efficiency and effectiveness in optimizing modularity. This approach uncovers 26 distinct geopolitical communities, as visualized in Fig. 2.

B. Step 2: Extraction of Key Recurrent Themes

Based on the thematic coherence and geopolitical alignment evident in their most viral tweets, communities are grouped into broader ideological camps, shown in Table II. Within each camp, we use topic modeling or message clustering to pinpoint their key recurring themes. This process systematically organizes content into key conceptual summaries, thereby preparing these neural themes for their formal transformation into symbolic logic structures.

We leverage Google’s NotebookLM [72] to extract key neural themes from each ideological camp. This involves loading the top 1000 most viral tweets per camp and applying a structured summarization prompt (Fig. 3). Google’s NotebookLM is an AI-powered research assistant designed to help users refine, organize, and interact with their documents and ideas. It distinguishes itself from traditional LLMs by being source-grounded, meaning it exclusively generates responses and insights based on the user-provided content. This approach

Objective: The attached dataset contains viral tweets discussing the global energy landscape, focusing on the energy needs of various countries and the role of energy corporations.

Instructions:

- 1) **Group tweets by country or regional actor**
(e.g., Russia, China, India, USA/America, Ukraine, UK, Palestine, Europe/EU, Canada, UAE, Qatar, Saudi Arabia, Egypt, Israel).
- 2) **Identify recurring themes**
Summarize key statements, claims, or frames that recur across tweets—whether verbatim or paraphrased.
- 3) **Classify framing polarity**
Determine whether each theme is **positively** or **negatively** framed based on whether it elevates or critiques a country, group, or policy.
- 4) **Group into narrative categories**
Organize themes into higher-level narrative types reflecting moral or rhetorical patterns (e.g., national security threat, corporate exploitation, energy independence, global cooperation, environmental justice).
- 5) **Focus on resonance and saturation**
Prioritize themes echoed across multiple users—those showing convergence and widespread recurrence.
- 6) **Produce a structured summary**
Present findings in a format that captures both the **country-level clustering** and the **narrative typologies**, with polarity and moral tone clearly indicated.

Fig. 3. Structured summarization prompt for presymbolic narrative analysis.

significantly reduces the risk of “hallucinations” and ensures transparency regarding where the information originates.

Key NotebookLM features supporting this task include.

- 1) *Source-Grounded Responses*: Outputs are restricted to the user-provided corpus and return “I don’t know” when unsupported.
- 2) *Inline Source Citations*: Responses include passage-level citations for traceability and auditability at scale; NotebookLM is used as a source-grounded summarizer, and any equivalent source-grounded LLM can substitute.

The prompt in Fig. 3 serves as an effective summarization mechanism. By grouping tweets by country, it contextualizes key recurring themes within geopolitical frames, anchoring them to stance-aware wedge patterns (Table I) such as US^+ , $THEM^-$, and causal forms such as $(CG^+) \Rightarrow (VAL^+)$, capturing claims such as “sustainable mining protects the environment” (Table I). Its emphasis on theme abstraction and cross-user resonance ensures robust, generalizable outputs.

Grouping themes into higher order narrative categories encourages abstraction beyond surface-level language, setting the stage for symbolic transformation. The focus on resonance and saturation ensures thematic robustness across users, enhancing generalizability. Ultimately, the prompt supports both lifting—by structuring neural patterns into interpretable rules—and lowering—by informing symbolic constraints within neural models, producing grounded outputs suited for transparent and trustworthy reasoning. Table III showcases a selection of key

recurring narrative themes and summaries, organized by geopolitical alignment and political stance.

C. Step 3: Symbolic Pattern Lifting from Neural Themes

Patterned narratives serve as the foundational programmable units in Human + AI text analysis, bridging the expressive variability of natural language with the formal precision required for machine reasoning. By reconciling variable syntax with fixed semantics, they enable scalable, interpretable translation of language into symbolic logic.

Each patterned narrative captures thematic alignments by linking topic-relevant noun phrases with polarized nouns or verbs, forming symbolic tuples that express core oppositions—such as threats to values, harms to common goods (often attributed to the out-group, e.g., *THEM*⁻), or benefits credited to the in-group (e.g., *US*⁺). Structured as conjunctions of disjunctions and aligned with the moral wedge templates in Table I, these patterns enable interpretable and programmable reasoning across diverse discourse contexts.

To derive these patterns, summaries and matched content from *NotebookLM* are systematically cross-referenced with the project’s codebook and wedge templates. This ensures that symbolic rules capture not only surface-level expressions, but also the deeper moral and political logic embedded within ideological perspectives. Few-shot prompting [73] enhances this process by embedding exemplar high-frequency pattern fragments into the symbolic lifting prompt (Fig. 4), guiding the model to produce outputs aligned with wedge templates.

The summarization prompt (Fig. 3) is then appended with the symbolic lifting prompt (Fig. 4) and exemplars, and re-run in *NotebookLM*, now equipped with structured symbolic scaffolding. The resulting patterned narratives, reflecting a selected subset of themes, are presented in Table III. Each patterned narrative includes an explainable summary, label, geopolitical camp, stance (pro or anti), and a corresponding Boolean expression. This mapping from variable surface text to stable symbolic structures enables the model to abstract both meaning and relational intent in a transparent, programmable form.

D. Steps 4 and 5: Pattern Lowering and Exception Handling

In essence, Step 3 enhances the algorithm’s capacity to translate validated sociopolitical hypotheses into symbolic patterns, thereby deepening its ability to identify and reason through salient narratives that cut across ideological divides.

These symbolic patterns are then applied to the dataset to generate pseudo-labels, which serve as supervisory signals for finetuning [74] and retraining a neural model—enabling it to learn from examples aligned with structured, rule-based logic.

As weaknesses in text analysis or generation emerge, the symbolic forms can be readily refined by humans to address exceptions, then reintegrated at Step 4—enabling incremental retraining without requiring full reinitialization.

Objective: Translate key recurring narrative themes into symbolic Boolean expressions—specifically, conjunctions of disjunctions over phrase-level predicates.

Instructions:

- 1) **Anchor each symbolic rule**
in a core narrative dichotomy—e.g., in-group moral praise vs. out-group blame (ACT), or causal links between threats/protections to common goods (CG) and values (VAL).
- 2) **Constrain the symbolic mapping**
using attached wedge templates and few-shot exemplars to ensure semantic and structural alignment.
- 3) **Incorporate relevant lexicons or phrase clusters**
tied to ideological actors, moral values, shared goods, and culturally resonant tropes in the attached codebook to ensure thematic specificity.
- 4) **Ensure each symbolic rule captures both surface features and latent function**
reflecting not just lexical similarity but the broader moral or political role of the theme in shaping affiliation, identity, blame, or justification.
- 5) **Produce a structured summary**
Produce compact, interpretable symbolic expression for each narrative that reflects the underlying narrative logic and are ready for downstream use in rule-based inference, causal modeling, or neurosymbolic integration.

Fig. 4. Structured prompt for lifting neural themes into symbolic logic patterns.

E. Writing Head: Programming a Campaigner AI

In today’s volatile and highly competitive global energy landscape, leadership demands more than abundant resources—it requires the strategic integration of advanced technologies, agile narrative control, and real-time geopolitical insight. Programmable AI is purpose-built to meet this challenge. It extracts actionable geopolitical intelligence and manages dynamic narrative ecosystems. By identifying emotionally resonant themes, high-impact influencers, and critical “trigger points,” it supports timely public opinion analysis and policy influence.

Research highlights this asymmetric influence: Linqia [75] reports that 7% of influencers control 90% of brand deals, while Baribi-Bartov et al. [76] find that fewer than 1% of Twitter users spread 80% of misinformation during the 2020 U.S. election. This underscores the strategic importance of targeting the most influential nodes in narrative networks.

The writing head extends these capabilities with structured, explainable, and programmable counter-narratives—such as rebuttals to claims of environmental harm—anchored in symbolic logic and aligned with morally and politically salient frames. These responses are not improvised but strategically crafted to achieve specific communicative objectives with transparency and accountability.

Though LLMs do not reason such as humans, they become powerful tools when grounded in symbolic logic, validated datasets, human wisdom, and judgment. Within this hybrid framework, LLMs function as narrative engines, capable of

TABLE III
PATTERNED NARRATIVE THEMES CATEGORIZED BY GEOPOLITICAL ALIGNMENT AND POLITICAL STANCE

Camp / Stance	Patterned Narrative	Theme Summary
CAMP: Sino-Eurasian Strategic Bloc		
anti-IOC/"Big Oil"	<i>Rollback of Green Initiatives and Deceptive Greenwashing:</i> {fundamentally reset, ditching, rollback, cut back, scam, deceptive, tactic, scrapped, halting, giving up} x {renewable, greenwash, green, climate, investment, green targets}	This theme criticizes IOCs for reducing or abandoning investments in renewable energy and for engaging in "greenwashing" - deceptively presenting themselves as environmentally friendly while continuing to focus on fossil fuels.
	<i>Fossil Fuel Dependence and Harm to the Planet:</i> {dirtiest fossil fuel, carbon, CO ₂ , fracking, emissions, harm, impact, cover-up} x {environment, environmental, greenhouse, green, net zero, fuel, gas, planet, climate-change, global warming}	Criticizes IOCs for their continued reliance on fossil fuels, particularly through practices like fracking, and their contribution to carbon emissions, greenhouse gases, and climate change. Allegations of IOCs knowing about the climate impacts of fossil fuels in the 1970s and lying about it are also present.
pro-IOC/"Big Oil"	<i>LNG and Natural Gas Development for Economic Growth:</i> {sustainable future, mini Plants, revolution, pipeline, growth, development, energy transition, investment} x {energy, gas, LNG, EU}	There is a narrative highlighting the role of natural gas and LNG development, often involving IOCs, in driving economic growth and energy transition.
	<i>Collaboration and Agreements for Gas Supply:</i> {sign deal, collaboration, securing contracts, met, business-friendly environment, cooperations, agreements, reexport} x {oil, gas pipeline, LNG, service contracts}	Agreements and collaborations involving IOCs for gas pipeline projects (Trans-Saharan Gas Pipeline), LNG supply to countries like Portugal, and development of gas fields are mentioned, often framed as positive steps for energy supply and international relations.
CAMP: Transatlantic Progressive Alliance		
anti-IOC/"Big Oil"	<i>Environmental Destruction and Harm:</i> {destruction, spillage, poison, suffering, pollution, toxic, damage, burns, burning} x {environment, environmental, health, communities, killing the planet, Mother Earth}	This recurrent theme focuses on the negative environmental and health consequences attributed to IOCs' operations, including oil spills, pollution, and the broader impact of emissions on the planet and local communities.
	<i>Fossil Fuel Dependence and Climate Crisis:</i> {cracker plant, carbon, CO ₂ , methane, fracking, emissions, cover-up, struggling to breathe} x {environment, greenhouse, net zero, fuel, gas, planet, climate, warming}	IOCs are perpetuating the climate crisis by their continued reliance on and expansion of fossil fuel extraction and use, despite awareness of the environmental harm, and that initiatives like "net zero" may be undermined by their actions. The term "natural gas" is often framed negatively as methane, a "climate super-pollutant".
pro-IOC Themes (Less Prominent and Often Countered)	<i>Economic Prosperity through Gas/Oil (Often Questioned):</i> {jobs, wealth, prosper} x {gas, oil}	Mentions of gas or oil leading to jobs and wealth are often countered by arguments about environmental costs, rising domestic prices due to exports, and the disproportionate benefits to the companies.
	<i>"Cleaner" Gas Narrative (Often Seen as Greenwashing):</i> {cleaner} x {LNG, gas, exports}	Claims about "cleaner LNG" are frequently juxtaposed with information highlighting that LNG can be worse than coal in terms of emissions.
CAMP: US & Western Conservative Coalition		
anti-Renewables/Green-Energy/Net-Zero	<i>Increased Energy Costs:</i> {higher rates, rising bills, most expensive electricity, unaffordable} x {renewable, green, Net Zero}	A prevalent negative asserting that the pursuit of renewable energy and Net-Zero policies leads to increased energy costs for consumers. Many tweets argue that green energy is inherently more expensive, requiring subsidies, and reliance on intermittent renewable sources drives up electricity prices.
	<i>Reliability of Renewable Energy Sources:</i> {unreliability, unreliable, can't power, fantasy crumbles, intermittency} x {renewable, green, solar, wind, AI, EVs, grid}	This theme criticizes the reliability of renewable energy sources like solar and wind, especially for demanding applications such as powering AI data centers and electric vehicles (EVs).
pro-LNG/Gas/Fracking	<i>Expansion of Gas and LNG:</i> {increased, production, exports, surging, potential, tremendous, open for business} x {gas, LNG, oil}	Focuses on the expansion of gas and LNG production and exports. The potential for new energy jobs due to increased LNG exports is noted. The surge in global demand for LNG is mentioned as well.
	<i>Positive Economic Outcomes:</i> {economic growth, job creation, reducing the trade deficit, lowering energy prices, win, very exciting, clean} x {energy, gas, LNG, oil}	This narrative links increased gas and LNG production and exports to positive economic outcomes such as job creation, reduced trade deficits, and lower energy prices for consumers.

Objective: Analyze the adversarial narrative labeled <insert malign narrative label here> from the <insert adversarial camp here>. Construct a strategically sound, ethically grounded counter-narrative using approved patterned narratives from <insert list of counter-narrative labels here>.

Instructions: Elevate in-group values, contrast with out-groups, and emphasize threats or protections to shared societal goods and personal values. Move beyond refutation—reframe the adversarial claim using symbolically coherent logic that aligns with the audience’s moral commitments.

Output Format:

- **Target Audiences** - origin of the adversarial narrative
- **Stance** - pro/anti position relative to the subject
- **Label** - concise counter-narrative theme
- **Patterned Narrative** - symbolic logic expression
- **Summary** - purpose and strategic intent.

Fig. 5. Structured prompt for counter-narrative generation.

organizing, contrasting, and responding to discourse patterns with evidentiary and moral coherence—transforming AI into a trustworthy partner in high-stakes communication.

The *Writing Head* supports ethical geopolitical analysis by contrasting narratives, extracting core arguments, and generating symbolic counter-narratives that clarify sacred values—advancing transparent, auditable reasoning under human oversight and responsible AI use.

Example: When anti-IOC narrative “*Environmental Destruction and Harm*” arises, the programmable writing head can use the prompt in Fig. 5 to generate a strategic, values-based response depicted in Fig. 6 utilizing an approved list of patterned counter-narratives.

As new narrative themes emerge, symbolic rules can be iteratively refined—without full system retraining—enabling adaptive and scalable narrative governance. Ultimately, this Human + AI campaigner architecture fuses LLMs’ pattern recognition capabilities with the moral clarity and programmatic transparency of human reasoning.

VI. QUANTITATIVE EVALUATION

To address the need for objective interpretability indicators and comparable evaluation metrics, we conduct a structured quantitative evaluation of the proposed programmable AI framework. Interpretability is operationalized through three measurable properties: 1) rule coverage, measuring the proportion of discourse that can be explicitly explained through symbolic patterns; 2) human–AI agreement, measuring the semantic fidelity of symbolic rule assignments relative to human interpretation; and 3) comparative predictive performance, benchmarking the symbolic narrative representation against purely neural clustering baselines.

These metrics quantify interpretability and comparability without altering neural weights, layers, or representations. Importantly, they evaluate properties of the symbolic rule interface

Target Audiences: Environmental Advocates and Climate-Conscious Stakeholders

Label: Powering Progress Responsibly

Stance: pro-LNG

Patterned Narrative:

{reliable, affordable, bridge fuel, clean, energy transition, lower energy prices, energy security, economic growth, job creation, Microsoft, Meta, AI, data centers, HUNGRY for power, 24/7 power, responsible} × {energy, gas, LNG, environment, global, prosperity}

Summary: In the face of rising global energy demands and the urgent need for sustainable solutions, natural gas—particularly LNG—offers a pragmatic bridge toward a low-carbon future. It is indispensable for powering critical infrastructure such as AI data centers that require uninterrupted, high-density power. As a cleaner, reliable, and scalable alternative to coal and oil, LNG supports decarbonization while ensuring economic growth, energy security, and affordability. This narrative reframes IOCs as essential partners in responsible energy stewardship, advancing shared prosperity and sustainability.

Fig. 6. Patterned counter-narrative generation.

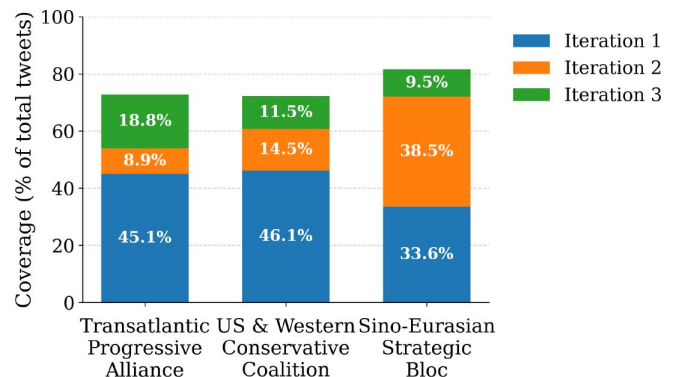


Fig. 7. Iterationwise symbolic rule coverage across ideological camps.

rather than modifications to neural model architecture: the proposed method externalizes neural representations into explicit symbolic logic structures rather than reorganizing internal neural weights.

A. Rule Coverage

We first evaluate how much of the discourse can be explained by symbolic narrative rules derived through the lifting process. Symbolic rules were generated using the structured summarization prompt (Fig. 3) followed by the symbolic lifting prompt (Fig. 4). The procedure was executed independently for each ideological camp identified in Table II.

For each camp, we analyzed the 1000 most-retweeted tweets. Narrative extraction and rule generation were performed iteratively. After each iteration, tweets already

TABLE IV
PRECISION, RECALL, AND F1 ACROSS CAMPS FOR TOP- k NARRATIVES ($k \in \{20, 40, 80\}$) AND CENTROID INITIALIZATION STRATEGIES

k	Init.	Transatlantic Progressive Alliance			US and Western Conservative Coalition			Sino-Eurasian Strategic Bloc		
		Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
20	Random	0.94	0.85	0.89	0.87	0.83	0.85	0.87	0.85	0.86
	Pattern + Summary	0.94	0.85	0.89	0.89	0.85	0.87	0.91	0.89	0.90
40	Random	0.91	0.85	0.88	0.86	0.82	0.84	0.90	0.88	0.89
	Pattern + Summary	0.95	0.85	0.90	0.86	0.82	0.84	0.94	0.91	0.92
80	Random	0.92	0.83	0.87	0.89	0.86	0.87	0.89	0.87	0.88
	Pattern + Summary	0.96	0.86	0.90	0.90	0.88	0.89	0.94	0.91	0.92

Note: Bold indicates the better-performing centroid initialization strategy.

matched by existing symbolic rules were removed before extracting additional patterned narratives from the remaining unmatched tweets. Rule coverage (RC) is defined as $RC = (\text{tweets matched by at least one symbolic rule}) / (\text{total tweets})$.

Fig. 7 presents the iterationwise coverage for each camp.

Across three iterations, the lifting process generated 113 patterned narratives for the *Transatlantic Progressive Alliance*, 103 patterned narratives for the *US and Western Conservative Coalition*, 90 patterned narratives for the *Sino-Eurasian Strategic Bloc*. After three iterations, the symbolic rule interface explained a substantial portion of the discourse within each camp (72.8%, 72.1%, and 81.6% respectively). Together, these results show that the symbolic narrative interface scales beyond anecdotes, providing substantial explanatory coverage within a few iterations and demonstrating that the narrative reasoning interface provides meaningful explanatory access to the underlying neural representations.

B. Human-AI Agreement (HAA)

To evaluate the reliability of the symbolic lifting process, we conducted a human validation study. A random sample of 600 rule-matched tweets was selected, with 200 samples drawn from each ideological camp. Each record contained the original tweet text, the matched symbolic narrative label and the narrative theme summary. Two independent human coders evaluated whether the symbolic rule correctly matched the tweet's narrative meaning. Interannotator agreement was measured using Cohen's κ . The resulting agreement score was: $\kappa = 0.84$, indicating almost perfect agreement.

Human disagreements were subsequently resolved through discussion to produce a consensus gold standard. Comparing symbolic rule assignments with this gold standard produced an overall rule precision of 90.33%. This high agreement demonstrates that symbolic lifting produces rule assignments that closely align with human interpretation of narrative framing.

C. Baseline: Pure Neural Versus Patterned Narratives

To evaluate whether symbolic narrative representations provide measurable advantages over purely neural representations, we compared the lifted patterned narratives with a neural clustering baseline. The neural baseline represents a purely

embedding-based narrative discovery pipeline without symbolic lifting or rule interfaces. Using Sentence-BERT [77] embeddings, we performed K-means clustering independently within each ideological camp. Clustering was evaluated at three granularities corresponding to the top- k most frequent narratives: $k \in \{20, 40, 80\}$.

For each configuration, we tested two centroid initialization strategies: *random initialization* (baseline neural clustering) and *pattern + summary initialization*, where cluster seeds incorporate symbolic pattern and narrative summary representations. Each cluster was assigned a dominant narrative label based on the most frequent narrative annotation among its member tweets.

For a given narrative N we compute.

- 1) *True Positives (TP)*: It tweets whose ground-truth narrative is N and whose cluster dominant label is also N
- 2) *False Positives (FP)*: It tweets assigned to clusters labeled N but whose true narrative differs
- 3) *False Negatives (FN)*: It tweets whose true narrative is N but assigned to clusters labeled differently

Precision, recall, and F1-score were then computed for each camp and configuration. Table IV summarizes the results. Across all configurations, clustering initialized with symbolic *pattern + summary representations* consistently improves alignment between clusters and narrative categories.

These results demonstrate that symbolic narrative representations not only preserve the semantic structure captured by neural embeddings but also introduce a stable, interpretable rule interface. Unlike purely neural clusters, which require post hoc interpretation, patterned narratives expose their internal logic through explicit Boolean structures that can be inspected, edited, and reused.

D. Interpretability Advantages

Beyond predictive performance, the programmable AI framework introduces structural interpretability advantages absent in purely neural models.

- 1) *Explicit rule interfaces* that expose narrative logic as Boolean symbolic patterns.
- 2) *Human-editable explanations* enabling direct correction of model behavior.
- 3) *Traceable reasoning paths* linking neural themes to symbolic narratives and downstream classifications.

Together, these results demonstrate that the proposed neuro-symbolic architecture achieves competitive predictive performance while providing measurable interpretability and human-verifiable reasoning.

VII. DISCUSSION AND CONCLUSION

We demonstrate that symbolic narrative patterning, the encoding of recurring themes as logical structures grounded in scientific hypotheses and human oversight, can transform opaque AI systems into interpretable and governable architectures. Through lifting (translating narrative text into symbolic logic) and lowering (embedding symbolic constraints back into neural models), the framework produces auditable reasoning paths aligned with human intent. At the heart of this framework is a programmable explanation: a symbolic scaffold that traces reasoning from data through hypotheses to conclusions, enabling direct inspection and refinement of the rules governing AI behavior.

We present a case study illustrating the end-to-end application of this methodology—from neural theme extraction, through symbolic pattern lifting, to knowledge-based constraint integration (lowering)—on real-world text data. Unlike neural pattern lifting, which seeks to interpret latent activations, our method operates on narrative content, transforming it into transparent, verifiable logic expressions that preserve interpretability and semantic alignment.

Our evaluation combines human review with quantitative validation to ensure both interpretability and reproducibility. Expert involvement is operationalized through a fixed protocol: 1) for each camp, we sample a predefined number of items (e.g., the 100 tweets nearest each cluster centroid for topical filtering, and 600 rule-matched tweets for rule validation); 2) two independent coders assess whether a rule correctly matches the tweet’s narrative meaning using a shared coding rubric; and 3) disagreements are resolved through documented adjudication to produce a consensus gold standard. This makes the human oversight step repeatable and auditable rather than ad hoc.

Across camps, symbolic lifting achieves 72.1%–81.6% rule coverage, 90.33% rule precision against the adjudicated gold standard, and interannotator agreement of $\kappa = 0.84$, indicating that the symbolic narrative interface provides broad explanatory coverage with high semantic reliability under a replicable expert-review procedure.

By grounding outputs in logic-based patterns, our system enables reliable validation, fine-tuning, and oversight. It ensures that models not only “say what they do” but also “do what they say,” supporting trustworthy and high-integrity reasoning across a wide range of social contexts—including social media, interviews, focus groups, and news sources.

While the proposed framework demonstrates strong interpretability and empirical performance, several considerations remain. The lifting–lowering pipeline may exhibit implementation-dependent variation across different LLM APIs and grounding mechanisms, and further large-scale benchmarking across domains, languages, and text modalities will strengthen claims of generalizability and scalability. In addition,

because symbolic rules are derived from observed discourse patterns, cross-domain validation and bias auditing are important to ensure robust and equitable deployment.

In conclusion, we argue that effective AI governance requires the independent ability to check, edit, and rigorously correct the symbolic foundations of an AI system’s reasoning—irrespective of the specific LLM upon which it operates. Without such capacity, all downstream outputs must be regarded as fundamentally unreliable. Our approach offers a principled path forward, making AI systems interpretable, steerable, and aligned with core human values, laying the foundation for building trust and ensuring the responsible use of AI in addressing critical societal challenges.

VIII. ETHICAL USE STATEMENT

This research advances symbolic reasoning in AI to enhance interpretability, accountability, and value alignment—particularly in sensitive areas such as geopolitical analysis and narrative intelligence. It is designed to augment human oversight, ensuring outputs remain transparent, auditable, and ethically grounded. The system is not intended for automated persuasion or disinformation. Its purpose is to support ethical, explainable decision-making through rigorously auditable symbolic patterning. Responsible use requires strict adherence to AI governance principles, individual autonomy, and safeguards against misuse in coercive narrative campaigns.

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